8B 1 TRENDS IN GROUPS 1 AND 2

SPECIFICATION REFERENCE
8.11 8.12

LEARNING OBJECTIVES

- Understand the reasons for the trend in ionisation energy down Groups 1 and 2.
- Understand the reasons for the trend in reactivity of the elements down Group 1 (Li to K) and Group 2 (Mg to Ba).

INTRODUCTION TO THE GROUP 1 AND 2 ELEMENTS

There are six elements in each of Groups 1 and 2. You are not likely to see samples of francium or radium, as all of their isotopes are radioactive. The other five elements in these groups look almost the same in appearance. You can describe all of them, when pure, as bright silvery solids. However, we have to keep them away from air to look like that. When exposed to air, they combine with oxygen to form oxides as surface layers, which makes them appear dull. **Fig A** shows the first five elements of Group 2 together.



▲ **fig A** From left to right: beryllium, magnesium, calcium, strontium and barium.

In this topic, we look at some trends in the properties of Group 1 and 2 elements.

TREND IN IONISATION ENERGY

We looked at ionisation energy in **Topic 2**. You may remember that it is a fundamental property that affects physical and chemical properties.

FIRST IONISATION ENERGIES IN GROUP 1

You should remember the definition of **first ionisation energy**. It is the energy required to remove an electron from each atom in one mole of atoms in the gaseous state. A general equation for this process, using M to represent an atom of any Group 1 element, is:

$$M(g) \rightarrow M^+(g) + e^-$$

Table A shows the metallic radius and the values of the first ionisation energies for the Group 1 elements.

ELEMENT	METALLIC RADIUS / nm	FIRST IONISATION ENERGY / kJ mol^{-1}
Lithium	0.152	519
Sodium	0.186	494
Potassium	0.231	418
Rubidium	0.244	402
Caesium	0.262	376

table A The metallic radii and the value of the first ionisation energies for the Group 1 elements.

The energy needed for ionisation is used to overcome the electrostatic attraction between the electron being removed and the protons in the nucleus.

FIRST AND SECOND IONISATION ENERGIES IN GROUP 2

Now we will consider the loss of two electrons from each atom, so we need to consider the **second ionisation energy.** This is the energy required to remove an electron from each singly charged ion in one mole of positive ions in the gaseous state. A general equation for this process is:

$$M^+(g) \rightarrow M^{2+}(g) + e^-$$

Table B shows the metallic radius and the values of the first and second ionisation energies for the Group 2 elements.

ELEMENT	METALLIC RADIUS / nm	IONISATION ENERGY / kJ mol^{-1}		
		FIRST	SECOND	1st + 2nd
Beryllium	0.112	900	1757	2657
Magnesium	0.160	738	1451	2189
Calcium	0.197	590	1145	1735
Strontium	0.215	550	1064	1614
Barium	0.224	503	965	1468

table B The metallic radii and the values of the first and second ionisation energies for the Group 2 elements.

The energy needed for ionisation is used to overcome the electrostatic attraction between the electron being removed and the protons in the nucleus.

FACTORS TO CONSIDER FOR GROUPS 1 AND 2

The factors to consider when explaining trends in ionisation energy are:

- the nuclear charge (or the number of protons in the nucleus)
- the orbital in which the electron exists
- the shielding effect (sometimes called the 'screening effect'). This is the repulsion between filled inner shells and the electron being removed.

In **Topic 2**, we considered the subshell (or sublevel) from which the electron is being removed. For the Group 1 and 2 elements this is not necessary because in their reactions the electrons are always removed from an s subshell.

You should be able to understand why the trend is a decrease down both Group 1 and Group 2.

- As the nuclear charge increases, so the force of attraction for the electron being removed also increases. This means an increase in ionisation energy down the group.

- As each quantum shell is added, energy of the outermost electrons increases. This means a decrease in ionisation energy down the group.
- As the number of filled inner shells increases, their force of repulsion on the electron being removed increases. This means a decrease in ionisation energy down the group.

You can see that the first factor causes an increase, but the second and third factors cause a decrease. The combined effect of the second and third factors outweighs the effect of the first factor. This means that there is a decrease down the group.

TREND IN REACTIVITY

We will look at the reactions of the elements of Groups 1 and 2 in **Topics 8B.2 to 8B.6**. In all these reactions, the element changes into either an M^+ ion or an M^{2+} ion, and there is a general increase in reactivity down the group. This can be explained by the decrease in energy needed to remove the electron from each atom of the element.

LEARNING TIP

When writing an equation for an ionisation, you should always include the state symbol (g) after each atom and ion.

CHECKPOINT

1. Write equations to represent the first ionisation of beryllium and the second ionisation of barium.

2. Explain fully why beryllium is less reactive than barium.

SUBJECT VOCABULARY

first ionisation energy the energy required to remove an electron from each atom in one mole of atoms in the gaseous state

second ionisation energy the energy required to remove an electron from each singly charged positive ion in one mole of positive ions in the gaseous state

8B 2 REACTIONS OF GROUP 1 ELEMENTS

SPECIFICATION REFERENCE
8.12 PART | 8.13 PART

LEARNING OBJECTIVES

- Know the reactions of the elements of Group 1 (Li to K) with oxygen, chlorine and water.

REACTIONS WITH OXYGEN

You know that the Group 1 metals are only shiny when kept out of air. When they are exposed to air, they tarnish. This means that they form a dull, dark layer on the surface as they are oxidised by oxygen in the air.

Group 1 metals can burn when heated in air or oxygen, but you would not normally do this in the laboratory because the reactions are extremely vigorous.

Many people are now aware of one problem to do with batteries containing lithium. Such batteries are common in mobile phones and laptops, but there have been a small number of cases where the battery has caught fire. **Fig A** shows a mobile phone on fire, caused by the lithium in the battery reacting rapidly with the oxygen in the air.



▲ **fig A** The lithium inside the battery of this mobile phone caught fire.

The general equation for all the reactions of Group 1 metals with the oxygen in the air is:

$$4M(s) + O_2(g) \rightarrow 2M_2O(s)$$

The products are oxides containing M^+ and O^{2-} ions.

Lithium is the least reactive Group 1 metal, and the reactivity of the metals increases down Group 1.

REACTIONS WITH CHLORINE

The Group 1 elements combine with chlorine when heated in chlorine gas. Just like the reactions with oxygen, the reactions with chlorine become more vigorous down the group, although this trend is harder to see than with the oxygen reactions.

The general equation for all of these reactions is:

$$2M(s) + Cl_2(g) \rightarrow 2MCl(s)$$

The products are chlorides containing M^+ and Cl^- ions.

REACTIONS WITH WATER

The reaction between lithium and water is vigorous. When a small piece of lithium is added to a beaker or trough of water, it reacts immediately and floats on the surface of the water. The fizzing is due to the formation of hydrogen gas and after a short while the lithium can no longer be seen. This is because it has reacted with the water to form a colourless solution of lithium hydroxide.

All the other Group 1 metals react in the same way, although the reactions become more vigorous down the group. When sodium is added to water, there is sometimes a flame, caused by the heat of the reaction igniting the hydrogen (**fig B**). When potassium is added to water, the hydrogen nearly always catches fire.

The general equation for all these reactions is:

$$2M(s) + 2H_2O(l) \rightarrow 2MOH(aq) + H_2(g)$$



▲ **fig B** The heat energy from the reaction between these pieces of sodium and water has ignited the hydrogen formed.

The products are hydrogen gas and hydroxides containing M^+ and OH^- ions.

The equation for the reaction with lithium is:

$$2Li(s) + 2H_2O(l) \rightarrow 2LiOH(aq) + H_2(g)$$

EXAM HINT

Make sure that you remember to use the correct symbols for reactants and products if asked to do so in an exam question.

REACTIONS OF OTHER GROUP 1 METALS

You are not required to know any of the reactions of rubidium or caesium. However, because you know that the trend is increasing reactivity down the group, you can predict that both rubidium and caesium are more reactive than potassium.

LEARNING TIP

Practise writing equations for reactions of Group 1 elements with oxygen, chlorine and water.

CHECKPOINT

1. Write an equation for each of these reactions:
   (a) lithium with oxygen
   (b) sodium with chlorine
   (c) potassium with water.

2. Soon after sodium is added to water, a colourless solution is formed. Explain why it is wrong to write that sodium is soluble in water.

8B 3 REACTIONS OF GROUP 2 ELEMENTS

SPECIFICATION REFERENCE
8.12 PART | 8.13 PART

LEARNING OBJECTIVES

- Know the reactions of the elements of Group 2 (Mg to Ba) with oxygen, chlorine and water.

REACTIONS WITH OXYGEN

You should be familiar with the reaction that occurs when magnesium burns in air – there is a very bright flame and the formation of a white solid.

Similar observations can be made when other Group 2 elements burn in air, but they are qualitatively different, that is, you should be able to recognise that when calcium, strontium and barium are heated in air, the reactions are more vigorous. However, this may be hard to see if the metals are not fresh samples. If the burning metal is placed in a gas jar of oxygen, then the same reaction occurs, although more vigorously.

For all these elements, the element needs to be heated for the reaction to start. However, even without heating, there is a slow reaction between the element and oxygen when the element is exposed to air. This forms a surface coating of oxide which helps to prevent the element from further reaction.

Barium is the most reactive. It is often stored under oil to keep it from reacting with oxygen and water vapour in the air.

The general equation for all these reactions is:

$$2M(s) + O_2(g) \rightarrow 2MO(s)$$

The products are oxides containing M^{2+} and O^{2-} ions.

(a)



(b)



▲ **fig A** (a) Burning magnesium in air. (b) Burning calcium in oxygen.

REACTIONS WITH CHLORINE

The Group 2 elements combine with chlorine when heated in the gas. Just like the reactions with oxygen, the reactions with chlorine become more vigorous down the group. However, this trend is harder to see than with the oxygen reactions. **Fig B** shows what magnesium burning in a flask of chlorine gas looks like.

The general equation for all of these reactions is:

$$M(s) + Cl_2(g) \rightarrow MCl_2(s)$$

The products are chlorides containing M^{2+} and Cl^- ions.



▲ **fig B** Magnesium burning in chlorine.

REACTIONS WITH WATER

The reaction between magnesium and water is very slow and does not proceed completely. Calcium, strontium and barium react with increasing vigour (i.e. reactivity increases down the group), which can be seen by the increase in effervescence.

In **fig C**, you can see that a piece of magnesium in water is covered with bubbles of hydrogen gas, but that the reaction is not very vigorous.

Fig D shows piece of calcium in water is also covered with bubbles of hydrogen gas, but the reaction is more vigorous.



▲ **fig C** Magnesium reacting very slowly with water.



▲ **fig D** Calcium reacting vigorously with water.

The general equation for all of these reactions is:

$$M(s) + 2H_2O(l) \rightarrow M(OH)_2(aq) + H_2(g)$$

The products are hydrogen gas and hydroxides containing M^{2+} and OH^- ions.

The equation for the reaction with calcium is:

$$Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(s) + H_2(g)$$

Calcium hydroxide is only slightly soluble in water, so the liquid in this experiment goes cloudy as a precipitate of calcium hydroxide forms.

The equation for the reaction with barium is:

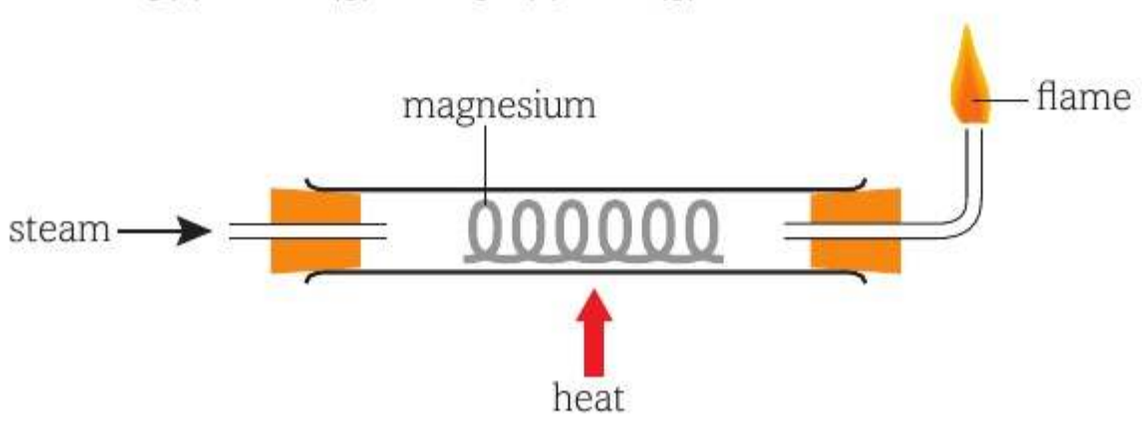
$$Ba(s) + 2H_2O(l) \rightarrow Ba(OH)_2(aq) + H_2(g)$$

Note the difference in the state symbol for the hydroxide in these equations, as barium hydroxide is soluble in water. You will learn more about the solubility of Group 2 hydroxides in the next topic.

MAGNESIUM AND STEAM

Magnesium reacts differently when heated in steam – it rapidly forms magnesium oxide (a white solid) and hydrogen gas in a vigorous reaction. The equation for this reaction is:

$$Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$$



The hydrogen formed is burned as it leaves the tube. This is for safety reasons, to prevent the escape of a highly flammable gas into the laboratory.

REACTIONS OF BERYLLIUM AND RADIUM

You are not required to know any of the reactions of beryllium and radium.

In a similar way to Group 1, the trend is increasing reactivity down the group, so you should be able to predict that beryllium is less reactive than magnesium, and that radium is more reactive than barium.

LEARNING TIP

Practise writing equations for reactions of Group 2 elements with oxygen, chlorine and water.

CHECKPOINT

1. Write an equation for each of these reactions:

(a) calcium with oxygen

(b) strontium with chlorine

(c) barium with water.

2. Suggest why it is not a good idea to use water to put out a fire involving burning magnesium.

8B 4 OXIDES AND HYDROXIDES IN GROUPS 1 AND 2

SPECIFICATION REFERENCE
8.14 8.15 8.19(i,ii)

LEARNING OBJECTIVES

- Know the reactions of the oxides of Group 1 and 2 elements with water and dilute acid, and their hydroxides with dilute acid.
- Know the trends in solubility of the hydroxides and sulfates of Group 2 elements.
- Know the reactions, including ionic equations where appropriate, for identifying carbonate and sulfate ions.

REACTIONS OF THE OXIDES WITH WATER

The Group 1 and 2 oxides are classed as **basic oxides**, which means that they can react with water to form alkalis. These reactions occur when the oxides are added to water. The only observation we can make is that the solids react to form colourless solutions. The general equations for these reactions are:

for Group 1 oxides: $M_2O(s) + H_2O(l) \rightarrow 2MOH(aq)$

for Group 2 oxides: $MO(s) + H_2O(l) \rightarrow M(OH)_2(aq)$

These equations can be simplified because there is no change to the M^+ or M^{2+} ion during the reactions.

$$O^{2-} + H_2O \rightarrow 2OH^-$$

This equation shows the formation of hydroxide ions, which is why the resulting solutions are alkaline.

TRENDS IN SOLUBILITY OF THE GROUP 2 HYDROXIDES

The pH value of the alkaline solution formed depends partly on the relative amounts of oxide and water, but is also affected by differences in the solubility of the hydroxides.

For example, when magnesium oxide reacts with water, the magnesium hydroxide formed has a very low solubility in water. The solubility of the Group 2 hydroxides increases down the group. Therefore, the maximum alkalinity (pH value) of the solutions formed also increases down the group.

TESTING FOR CARBON DIOXIDE

You may remember from your previous study of chemistry that limewater is used to test for carbon dioxide. It goes cloudy (or milky) as a white precipitate forms. Limewater is the name used for a saturated aqueous solution of calcium hydroxide. Carbon dioxide reacts to form calcium carbonate, the white precipitate, which is insoluble in water. The equation for the reaction is:

$$CO_2(g) + Ca(OH)_2(aq) \rightarrow CaCO_3(s) + H_2O(l)$$

As carbon dioxide is bubbled through limewater, the amount of precipitate increases (**fig A**).



▲ **fig A** As carbon dioxide is bubbled through limewater, the amount of precipitate increases.

MILK OF MAGNESIA

For over a century, a suspension of magnesium hydroxide in water has been sold as an indigestion remedy called milk of magnesia. A bottle of this contains a saturated solution of magnesium hydroxide mixed with extra solid magnesium hydroxide, which acts as an antacid.

The human stomach contains hydrochloric acid that is needed to digest food, but sometimes there is too much acid and the person develops symptoms of indigestion. Taking milk of magnesia neutralises some of the hydrochloric acid and relieves the symptoms. The equation for the reaction is:

$$Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$$

Although hydroxide ions attack human tissue, the very low solubility of magnesium hydroxide means that the concentration of OH^- ions in the medicine is also very low and is not a risk to health.



▲ **fig B** A bottle of milk of magnesia.

REACTIONS OF THE OXIDES AND HYDROXIDES WITH ACIDS

All of the Group 1 and 2 oxides and hydroxides react with acids to form salts and water. These reactions can be described as neutralisation reactions.

During the reactions, the only observations to be made are that a white solid reacts to form a colourless solution. The reactions are exothermic, so you may use some of them in experiments to measure energy changes.

Here are some sample equations:

$$Na_2O + H_2SO_4 \rightarrow Na_2SO_4 + H_2O$$

$$CaO + 2HNO_3 \rightarrow Ca(NO_3)_2 + H_2O$$

$$KOH + HCl \rightarrow KCl + H_2O$$

$$Ba(OH)_2 + 2HCl \rightarrow BaCl_2 + 2H_2O$$

USE IN AGRICULTURE

For centuries, farmers have used lime to control soil acidity so that a greater yield of crops can be obtained.



▲ **fig C** Lime being spread on a field.

Lime is mostly calcium hydroxide (obtained from limestone, which is calcium carbonate), and neutralises excess acidity in the soil. Using nitric acid to represent the acid in soil, the equation for this reaction is:

$$Ca(OH)_2 + 2HNO_3 \rightarrow Ca(NO_3)_2 + 2H_2O$$

TRENDS IN SOLUBILITY OF THE GROUP 2 SULFATES

All Group 2 nitrates and chlorides are soluble, but the solubility of Group 2 sulfates decreases down the group.

- Magnesium sulfate is classed as soluble.
- Calcium sulfate is slightly soluble.
- Strontium sulfate and barium sulfate are insoluble.

You do not have to understand the reasons for this trend, but you do need to know how the very low solubility of barium sulfate is used in a test for sulfate ions in solution.

TESTING FOR SULFATE IONS

The presence of sulfate ions in an aqueous solution can be shown by adding a solution containing barium ions (usually barium chloride or barium nitrate). Any sulfate ions in the solution will react with the added barium ions to form a white precipitate of barium sulfate. The ionic equation for this reaction is:

$$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$$

There are other anions that could also form a white precipitate with barium ions, especially carbonate ions, so in the test there must be H^+ ions present to prevent barium carbonate from forming as a white precipitate. Dilute nitric acid or dilute hydrochloric acid is therefore added as part of the test.

As an example, here is how to test for the presence of sulfate ions in a solution of sodium sulfate.

- Add dilute nitric acid and barium nitrate solution to the sodium sulfate solution.
- A white precipitate forms.

The equation for the reaction is

$$Ba(NO_3)_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaNO_3(aq)$$

BARIUM MEALS

Solutions containing barium ions are poisonous to humans, however, barium sulfate is used in hospitals, where patients are sometimes given a barium 'meal'. This 'meal' contains barium sulfate, which is not poisonous because it is insoluble – although it contains barium ions, these ions are not free to move. Although bones can be seen clearly on X-rays, soft tissues cannot. If the patient has a barium meal before an X-ray, these soft tissues will show up more clearly because of the dense white solid.



▲ **fig D** How a barium meal can help show up soft tissues on an X-ray.

LEARNING TIP

There is no trend in reactivity with water for the Group 1 and 2 oxides, because they already contain metal ions, not metal atoms.

CHECKPOINT

1. Limewater is used to test for carbon dioxide. Why should limewater not be left exposed to air before using it in this test?
2. Why is an acid added when using barium chloride or barium nitrate solution to test for sulfate ions?

SUBJECT VOCABULARY

basic oxides oxides of metals that react with water to form metal hydroxides, and with acids to form salts and water

8B 5 THERMAL STABILITY OF COMPOUNDS IN GROUPS 1 AND 2

SPECIFICATION REFERENCE
8.16 8.18(i)

LEARNING OBJECTIVES

- Understand the reasons for the trends in thermal stability of the nitrates and the carbonates of the elements in Groups 1 and 2 in terms of the size and charge of the cations involved.
- Know experimental procedures used to show patterns in the thermal decomposition of Group 1 and 2 nitrates and carbonates.

FACTORS AFFECTING THERMAL STABILITY

Thermal stability is a term that indicates how stable a compound is when it is heated. Does it not decompose at all (very thermally stable), or does it decompose as much as possible (not at all thermally stable), or somewhere in between?

You are familiar with ionic bonding in compounds and you know that the bond strength is responsible for physical properties such as the melting temperature. For example, the high melting temperature of sodium chloride (NaCl) can be explained in terms of the strong electrostatic forces of attraction between the large numbers of oppositely charged Na^+ and Cl^- ions. When sodium chloride melts, the ions change from being regularly arranged in a giant lattice to moving freely in a liquid. There is no decomposition occurring in this change of state.

The situation with Group 2 nitrates and carbonates is very different compared with a Group 1 chloride such as sodium chloride. There are three reasons for this.

1 The charge on a Group 2 cation is double that on a Group 1 cation (e.g. Ca^{2+} compared with Na^+).
2 The size (ionic radius) of a Group 2 cation is smaller than that of the Group 1 cation in the same period.
3 The nitrate (NO_3^-) and carbonate (CO_3^{2-}) anions are more complex than the Cl^- ion.

These differences mean that when Group 2 nitrates and carbonates are heated, they do not melt. Instead, they decompose. We need to look carefully at these factors to understand why this is.

- The larger, more complex nitrate ion can change into the smaller, more stable nitrite ion (NO_2^-) or oxide ion (O^{2-}) by decomposing and releasing oxygen gas and/or nitrogen dioxide gas.
- The larger, more complex carbonate ion can change into the smaller, more stable oxide ion (O^{2-}) by decomposing and releasing carbon dioxide gas, CO_2.
- The stabilities of the nitrate and carbonate anions are influenced by the charge and size of the cations present. Smaller and more highly charged cations affect these anions more.

Table A shows the charge and radius for each of the ions in Groups 1 and 2.

GROUP 1			GROUP 2		
ELEMENT	CHARGE ON ION	IONIC RADIUS / nm	ELEMENT	CHARGE ON ION	IONIC RADIUS / nm
Lithium	+1	0.074	Beryllium	+2	0.027
Sodium	+1	0.102	Magnesium	+2	0.072
Potassium	+1	0.138	Calcium	+2	0.100
Rubidium	+1	0.149	Strontium	+2	0.113
Caesium	+1	0.170	Barium	+2	0.136

table A

You can see that the cation with the greatest influence (biggest charge and smallest size) on an anion is Be^{2+}, and the one with the least influence (smallest charge and largest size) is Cs^+.

THERMAL STABILITY OF NITRATES

All of the nitrates of the Group 1 and Group 2 elements are white solids. When they are heated, they all decompose to nitrites or oxides, and give off nitrogen dioxide (brown fumes) and/or oxygen. If the nitrate contains water of crystallisation, then steam will also be observed.

If no brown fumes are observed, this indicates a lesser decomposition. This decomposition can be represented by this word equation:

metal nitrate → metal nitrite + oxygen

Oxygen gas cannot be observed as it is colourless. It can be detected using a glowing spill or splint, which relights if oxygen is present.

Nitrates and nitrites are sometimes differentiated by using oxidation numbers – nitrate(V) for nitrate, and nitrate(III) for nitrite.

If brown fumes are observed, this indicates a greater decomposition that can be represented by this word equation:

metal nitrate → metal oxide + nitrogen dioxide + oxygen

Table B shows typical observations obtained by heating samples of nitrates in test tubes over a Bunsen flame.

GROUP 1 NITRATE		GROUP 2 NITRATE	
NAME	RESULT	NAME	RESULT
Lithium nitrate	brown fumes	Beryllium nitrate	brown fumes
Sodium nitrate	no brown fumes	Magnesium nitrate	brown fumes
Potassium nitrate	no brown fumes	Calcium nitrate	brown fumes
Rubidium nitrate	no brown fumes	Strontium nitrate	brown fumes
Caesium nitrate	no brown fumes	Barium nitrate	brown fumes

table B

Table C shows what happens, in terms of decomposition, when samples of nitrates are heated in test tubes over a Bunsen flame.

GROUP 1 NITRATE		GROUP 2 NITRATE	
NAME	RESULT	NAME	RESULT
Lithium nitrate	greater decomposition	Beryllium nitrate	greater decomposition
Sodium nitrate	lesser decomposition	Magnesium nitrate	greater decomposition
Potassium nitrate	lesser decomposition	Calcium nitrate	greater decomposition
Rubidium nitrate	lesser decomposition	Strontium nitrate	greater decomposition
Caesium nitrate	lesser decomposition	Barium nitrate	greater decomposition

table C

You can see that the greater decomposition occurs when:

- the cation has a 2+ charge (all of the Group 2 nitrates)
- the cation has a 1+ charge and is also the smallest Group 1 cation.

Here are some sample equations for the reactions that occur:

$4LiNO_3 \rightarrow 2Li_2O + 4NO_2 + O_2$
(lithium nitrate – the only Group 1 nitrate that decomposes in this way)

$2NaNO_3 \rightarrow 2NaNO_2 + O_2$
(all other Group 1 nitrates decompose in this way)

$2Be(NO_3)_2 \rightarrow 2BeO + 4NO_2 + O_2$
(all Group 2 nitrates decompose in this way)

EXAM HINT

Previous exam questions suggest that many students are poor at reproducing these equations under exam conditions so be sure to learn them.

EXAM HINT

Note that these reactions involving decompositions of nitrates are redox reactions. Check by assigning oxidation numbers to nitrogen and oxygen in each reaction.

THERMAL STABILITY OF CARBONATES

All of the carbonates of the Group 1 and Group 2 elements are white solids. When they are heated, they either do not decompose, or decompose to oxides and give off carbon dioxide.

As the gas given off is colourless and the carbonate and oxide are both white solids, there are no observations that can be made.

Table D shows what happens when samples of carbonates are heated in test tubes over a Bunsen flame.

GROUP 1 CARBONATE		GROUP 2 CARBONATE	
NAME	RESULT	NAME	RESULT
Lithium carbonate	decomposition	Beryllium carbonate	decomposition
Sodium carbonate	no decomposition	Magnesium carbonate	decomposition
Potassium carbonate	no decomposition	Calcium carbonate	decomposition
Rubidium carbonate	no decomposition	Strontium carbonate	decomposition
Caesium carbonate	no decomposition	Barium carbonate	decomposition

table D Results, in terms of decomposition, when samples of carbonates are heated.

You can see that the pattern is similar to that of the nitrates. The lithium compound and all of the Group 2 compounds behave differently from the other Group 1 compounds.

Lithium carbonate decomposes at lower temperatures than the other Group 1 carbonates:

$$Li_2CO_3 \rightarrow Li_2O + CO_2$$

Other Group 1 carbonates do not decompose on heating, except at very high temperatures.

All Group 2 carbonates decompose in the same way, but with increasing difficulty down the group. A typical equation for one of these decompositions is:

$$CaCO_3 \rightarrow CaO + CO_2$$

You can see a similar pattern as with the nitrates. Decomposition occurs when:

- the cation has a 2+ charge (all of the Group 2 carbonates)
- the cation has a 1+ charge and is also the smallest Group 1 cation (only lithium carbonate).



▲ **fig A** Calcium oxide forms when calcium carbonate (limestone) is heated strongly. If the strong heating is continued, there is no further chemical change, but a very bright glow is seen. This is the origin of 'limelight' – it was formerly used in theatre lighting.

LEARNING TIP

Look for the similar patterns in the thermal stability of Group 1 and Group 2 nitrates and carbonates. Although the reactions are different, the lithium compound behaves like the Group 2 compounds and not like the other Group 1 compounds.

CHECKPOINT

1. What observations would be made when these compounds are heated in a test tube over a Bunsen flame?
   (a) calcium nitrate
   (b) sodium carbonate.
2. Write a chemical equation for each of these reactions.
   (a) the decomposition of potassium nitrate
   (b) the decomposition of strontium carbonate.

SUBJECT VOCABULARY

thermal stability a measure of the extent to which a compound decomposes when heated

8B 6 FLAME TESTS AND THE TEST FOR AMMONIUM IONS

SPECIFICATION REFERENCE
8.17 8.18(ii) 8.19(iii)

LEARNING OBJECTIVES

- Know experimental procedures used to show flame colours in compounds of Group 1 and 2 elements.
- Understand the formation of characteristic flame colours by Group 1 and 2 compounds in terms of electron transitions.
- Know how to identify ammonium ions.

INTRODUCTION TO FLAME TESTS

A flame test seems a very simple chemical test to identify the presence of a cation in a compound. However, it is important to know how to do one carefully, to know how to interpret the results, and to understand how the test works.

A flame test result can indicate the presence of some metals (in the form of cations) in Groups 1 and 2 of the Periodic Table. It does not work for all of them.

HOW TO DO A FLAME TEST

We will assume that a small quantity of a solid is available in a glass dish. This is how to do a flame test.

- Wear safety glasses and a lab coat. Within a fume cupboard, light a Bunsen burner.
- Using a dropper, add a few drops of concentrated hydrochloric acid to the solid and mix together so that the metal compound begins to dissolve. (One reason for using hydrochloric acid is to convert any metal compound to a chloride – chlorides are more volatile than other salts so are more likely to give better results.)
- Dip a clean metal wire (platinum or nichrome) or silica rod into the mixture to obtain a sample of the compound.
- Hold the end of the wire or rod in the flame and observe the colour.



▲ **fig A** A platinum wire loop with a sample of a strontium compound being held in a Bunsen flame.

PROBLEMS WITH A FLAME TEST

There are two main problems with a flame test.

- Many compounds contain small amounts of sodium compounds as impurities, so the intense colour of sodium can hide other colours.
- Describing colours with words is subjective – people have different levels of colour vision, and a word description of a colour may mean different colours to different people.

COLOUR DESCRIPTIONS

Some traditional descriptions of colours are problematic. For example, what is the colour 'brick red'? It may depend on the bricks used in a particular location, and these are not the same throughout the world.

Another example is the traditional colour description used for potassium. It is lilac, but can you tell the difference between lilac and lavender, magenta, mauve, pink, plum, puce, purple and violet?

Table A shows some traditional colour descriptions obtained when Group 1 and 2 compounds are tested in this way.

METAL CATION	FORMULA	COLOUR
Lithium	Li^+	red
Sodium	Na^+	yellow/orange
Potassium	K^+	lilac
Rubidium	Rb^+	red/purple
Caesium	Cs^+	blue/violet
Beryllium	Be^{2+}	(no colour)
Magnesium	Mg^{2+}	(no colour)
Calcium	Ca^{2+}	(brick) red
Strontium	Sr^{2+}	(crimson) red
Barium	Ba^{2+}	(apple) green

table A Colour descriptions obtained when Group 1 and 2 compounds undergo a flame test.

There are also some metal cations with characteristic flame colours that are not in Groups 1 or 2. For example, copper compounds produce a blue-green colour in a flame test.

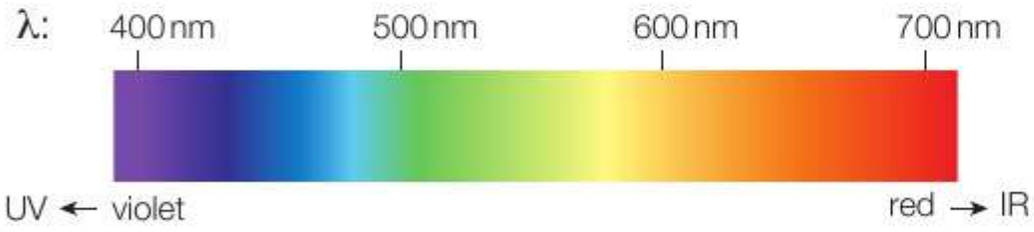
WHAT CAUSES THE COLOURS IN FLAME TESTS?

The simple answer is electron transitions. However, we need a more complete explanation than this.

You know from **Topic 2** that electrons occupy orbitals in specific energy levels in an atom. These are often represented using electron configurations such as 2.8.1 (for sodium).

Electrons can absorb energy and move to higher energy levels. Sometimes the term 'ground state' is used to describe an atom with all its electrons in their lowest possible energy levels. If an electron moves to a higher energy level, then the new situation can be described as an 'excited state'. This movement of an electron to an excited state occurs during a flame test.

However, this movement is immediately followed by the return of the electron to its ground state, which releases energy. If this energy corresponds to radiation in the visible light spectrum, then a characteristic colour appears – this is the flame test colour.



▲ **fig B** The wavelengths of the spectrum of visible light.

For example, the visible spectrum covers the part of the electromagnetic spectrum in the wavelength range 400–700 nm. The electron transition in sodium corresponds to a wavelength of about 590 nm, which is in the yellow-orange part of the spectrum, so this is the colour of a sodium compound in a flame test. The electron transition in magnesium corresponds to a wavelength outside the visible spectrum, so there is no flame colour for magnesium.



▲ **fig C** The yellow-orange colour of sodium street lights is the same as the colour of sodium in a flame test.

LEARNING TIP

Remember that doing a flame test on a magnesium compound has nothing to do with burning magnesium metal in a Bunsen flame. When magnesium metal burns, it changes from Mg atoms to Mg^{2+} ions. Magnesium compounds already contain Mg^{2+} ions.

THE TEST FOR AMMONIUM IONS

One cation that does not give a colour in a flame test is the ammonium ion. The usual test for ammonium ions in a solid or solution is to add sodium hydroxide solution and warm the mixture. The addition of sodium hydroxide causes this reaction:

$$NH_4^+ + OH^- \rightarrow NH_3 + H_2O$$

and the warming releases ammonia gas. Ammonia can be recognised by its smell, but you can use damp litmus paper as a simple chemical test, which turns blue (ammonia is the only common alkaline gas). Alternatively, hydrogen chloride gas (from concentrated hydrochloric acid) reacts with ammonia to form white smoke of ammonium chloride:

$$NH_3 + HCl \rightarrow NH_4Cl$$

EXAM HINT

If you are asked for a safety precaution for these reactions, it is worth saying that they should be carried out in a fume cupboard.

CHECKPOINT

1. Why is concentrated hydrochloric acid used in flame tests?

2. Explain why barium compounds give a characteristic flame colour but magnesium compounds do not.